CHROME-pred | AI-Assisted HPLC Method Development

Chromatography AI System v3.0 — Comprehensive Report

Field	Value
Report ID	RPT-20260526151056-e3d836
Generated	2026-05-26T15:10:56.002007
Compound Name	N/A
SMILES	CC(=O)OC1=CC=CC=C1C(=O)O
Project	N/A
Status	Success
Processing Time	0.05 s
Overall Confidence	0.32

Module Confidence Scores

Module	Score	Status	Rationale
Physicochemical	0.00	Low	
HSM	0.65	Moderate	HSM descriptors (eta', sigma', beta', alpha', kappa') are calculated to capture hydrophol
Buffer	0.01	Low	Buffer recommendations are based on pKa matching, buffering range, and compatibility
Solvent	0.88	High	Solvent scoring considers polarity, hydrogen-bonding character, and miscibility to priorit
Column	0.07	Low	Column ranking uses HSM-derived retention parameters and known selectivity profiles
Overall	0.32	Low	

Section 1 — Compound Identification & Physicochemical Profile

1.1 Compound Information

Property	Value
Compound Name	N/A
SMILES	CC(=O)OC1=CC=CC=C1C(=O)O
Molecular Formula	N/A
Molecular Weight	N/A g/mol
Exact Mass	N/A Da
Project	N/A

1.2 Physicochemical Properties

Descriptor	Value	Unit	Threshold	Reference
LogP (Wildman-Crippen)	N/A		–2 to 5	Wildman & Crippen 1999
LogD (pH 2.0)	N/A			Henderson-Hasselbalch
LogD (pH 5.0)	N/A			
LogD (pH 7.4)	N/A			Physiological pH
LogD (pH 9.0)	N/A			
LogD (pH 11.0)	N/A			
TPSA	N/A	$Å^2$	≤140	Ertl et al. 2000
H-Bond Donors	N/A		≤5	Lipinski 1997
H-Bond Acceptors	N/A		≤10	
Rotatable Bonds	N/A		≤10	Veber 2002
Aromatic Rings	N/A			
Fraction Csp^3	N/A			

1.3 Drug-Likeness

Filter	Status	Threshold	Reference
Lipinski Violations	N/A	≤1 acceptable	Lipinski 1997
Ghose Filter	N/A	MW 160-480	Ghose 1999
Veber Filter	N/A	Rot ≤10, TPSA ≤140	Veber 2002
Muegge Filter	N/A	MW 200-600	Muegge 2001
QED Score	N/A	>0.5 favourable	Bickerton 2012

1.4 Ionization Profile

Parameter	Value
Ionization Type	N/A
pKa Acidic (min)	N/A
pKa Basic (max)	N/A
Isoelectric Point	N/A
Formal Charge	N/A

1.5 Solubility & Permeability

Parameter	Value	Unit	Reference
LogS (ESOL)	N/A	log mol/L	Delaney 2004
Intrinsic Solubility	N/A	mg/mL	
Caco-2 Permeability	N/A	×10■■ cm/s	PAMPA
MDCK Permeability	N/A	×10■■ cm/s	

Section 2 — HSM Descriptors

2.1 Descriptor Values (Snyder et al., 2004)

Descriptor	Symbol	Value	Interpretation	Reference
Hydrophobicity	η′			Marchand 2008
Steric Resistance	σ′			Marchand 2005
H-Bond Basicity	β′			Abraham 1993
H-Bond Acidity	α′			Abraham 1993
Cationic Charge	κ′			Dolan 2004
Descriptor	**Symbol**	**Value**	**Interpretation**	**Reference**
Hydrophobicity	η′	1.944	High	Marchand 2008
Steric Resistance	σ′	0.467	High	Marchand 2005
H-Bond Basicity	β′	1.000	Strong	Abraham 1993
H-Bond Acidity	α′	0.800	Strong	Abraham 1993
Cationic Charge	κ′	0.000	Min.	Dolan 2004

2.2 Estimation Formulae

η′:

```
η′ = 0.8 × LogP + 0.3 + 0.2 × (MR/15)
```

σ′:

```
σ′ = 0.5 − 0.15 × ShapeIndex + 0.05 × ln(Rot+1) − 0.02 × Rings
```

β′:

```
β′ = min(1, 1.0×N_alip_N + 0.6×N_arom_N + 0.5×N_carbonyl + 0.3×N_ether)
```

α′:

```
α′ = min(1, 0.8×COOH + 0.7×ArOH + 0.5×Amide_NH + 0.4×AlkOH)
```

κ′:

```
κ′ = N_quaternary + f(pH)×(N■+N■+N■) + g(pH)×N_arom_N
```

Reference: Snyder et al. (2004) J. Chromatogr. A 1060:77-116

Section 3 — Column Selection & Ranking

3.1 Scoring Algorithm

Applied scoring function: **Basic**

```
Score = 2.0×η′×H − 3.0×β′×A − 2.0×κ′×C [Dolan et al. 2004]
```

3.2 Top Column Recommendations

Rank	Column	Manufacturer	H	S	A	B	C	Score
1	Inertsil ODS-EP	GL Sciences	0.800	0.060	-1.520	0.050	-0.070	7.037
2	Zorbax Bonus RP	Agilent	0.650	0.100	-1.040	0.370	-1.100	5.132
3	ZirChrom-PBD	ZirChrom	1.280	0.150	-0.380	-0.070	2.180	5.103
4	ProntoSIL 120 C18 ace-EPS	Bischoff	0.770	0.040	-0.590	0.220	0.040	4.154
5	Acclaim PolarAdvantage II	Dionex	0.740	0.010	-0.550	0.210	0.670	3.941

3.4 Column Equivalence (Fs Factor)

```
Fs = √[(12.5×ΔH)² + (100×ΔS)² + (30×ΔA)² + (143×ΔB)² + (83×ΔC)²]
```

Fs Value	Interpretation
Fs ≤ 3	Equivalent — interchangeable
3 < Fs < 5	Similar — minor differences possible
Fs ≥ 5	Different selectivity — re-validation required

Section 4 — Solvent Selection

4.1 Scoring System (7 Rules)

Rule	Weight	Description	Reference
1. LogP Matching	25%	ΔLogP 2.0–3.5 optimal	Valkó 2004
2. HBD/HBA	20%	C_score < 0.3	Vitha & Carr 2006
3. Polarity	15%	|ΔP'| ≤ 2.0	Snyder 1974
4. Kamlet-Taft	15%	D < 0.5	Carr 1993
5. Viscosity	10%	Low backpressure	Li & Carr 1997
6. UV Transparency	10%	Margin ≥ 20 nm	Dolan 1999
7. pH Stability	5%	Within column range	Neue 1997

4.2 Top Solvent Recommendations

Rank	Solvent	Overall Score	Notes
1	Ethanol	87.6	
2	Methanol	86.4	
3	Acetonitrile	79.4	

4.3 Initial Mobile Phase Composition

Parameter	Value
Organic Modifier %	N/A%
Aqueous %	N/A%
Rationale	N/A

Section 5 — Buffer Selection

5.1 10-Rule Scoring System

Rule	Weight	Description
1. pKa Matching	20%	Buffer ±1 pH unit of pKa (Goldberg 2002)
2. UV Transparency	15%	λ_det > λ_cutoff + 10 nm
3. Organic Solubility	12%	Soluble throughout gradient
4. Buffer Capacity	10%	$\beta \geq 0.01$
5. MS Volatility	15%	Volatile buffers for LC-MS (Kebarle 1993)
6. Temperature Dep.	8%	pKa(T) correction
7. Chemical Reactivity	10%	Avoid analyte reactions
8. Ionic Strength	3%	I = 20–50 mM
9. Storage Stability	2%	≤7 days at RT
10. Metal Complexation	5%	Avoid chelators

5.2 Top Buffer Recommendations

Rank	Buffer	pKa	Score	Notes
1	HEPES	7.48	0.849	—
2	Phosphate	7.20	0.709	Failed: Rule 3: Organic Solubility (max 65% ACN); Failed: Rule 7: C
3	Bicarbonate	6.33	0.707	Failed: Rule 3: Organic Solubility (max 70% ACN)
4	MES	6.10	0.643	—
5	Formate	3.75	0.640	Failed: Rule 1: Buffer pKa Matching (±0.5 optimal, ±1.0 effective)

5.3 Detailed Scoring — HEPES

Rule	Score	Status
pKa Matching	0.000	Fail
UV Transparency	0.000	Fail
Organic Solubility	0.000	Fail
Buffer Capacity	0.000	Fail
MS Volatility	0.000	Fail
Temp. Dep.	0.000	Fail
Reactivity	0.000	Fail
Ionic Strength	0.000	Fail
Storage	0.000	Fail
Metals	0.000	Fail

Section 6 — Gradient Program Optimization

6.1 Gradient Parameters

Parameter	Value	Reference
Total Runtime	N/A min	—
Peak Capacity	N/A	—
Min. Resolution	N/A	ICH Q2(R2)
Column Length	150 mm	—
Column ID	4.6 mm	—
Particle Size	3.5 µm	—

6.2 Gradient Programme

Time (min)	%B	Type	Description
0.00	5.0	Initial	—
10.00	95.0	End	5→95%B

6.3 LSS Theory Rationale

b = `b = S × Δϕ × t■ / t_G [Snyder & Dolan 2007]`

t_R = `t_R = t■ + (t■/b) × ln(1 + b × k■) [Neue 1997]`

S = `S ≈ 0.48 × MW^0.44 × (1 + 0.1 × (LogP − 3))`

Section 7 — Confidence Assessment & Warnings

7.1 Module Confidence Scores

Module	Score	Status	Rationale
Physicochemical	0.00	Low	
HSM	0.65	Moderate	HSM descriptors (eta', sigma', beta', alpha', kappa') are calculated to capture hydrophol
Buffer	0.01	Low	Buffer recommendations are based on pKa matching, buffering range, and compatibility
Solvent	0.88	High	Solvent scoring considers polarity, hydrogen-bonding character, and miscibility to priorit
Column	0.07	Low	Column ranking uses HSM-derived retention parameters and known selectivity profiles
OVERALL	0.32	Low	

7.2 Interpretation Scale

Score Range	Interpretation	Recommendation
> 0.80	High confidence	Method works well, minimal optimisation
0.50–0.80	Moderate confidence	Some optimisation required
< 0.50	Low confidence	Significant optimisation + validation needed

7.3 Warnings & Limitations

■ PhysChem: module 'rdkit.Chem' has no attribute 'MolToInchi'

■ Gradient: 'float' object is not iterable

■ HSM descriptors are empirical estimates — experimental validation recommended.

■ pKa predictions are fragment-based approximations (±1 pH unit accuracy).

■ Gradient predictions assume ideal linear solvent strength (LSS) behaviour.

Section 8 — References (Vancouver Style)

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18. Snyder LR, Kirkland JJ, Dolan JW. Intro. to Modern LC. 3rd ed. Wiley; 2010.
19. USP PQRI Column Equivalence Database. https://apps.usp.org/app/USPNF/columnsDB.html

Section 9 — Regulatory Alignment (ICH / USP)

Guideline	Applicability	Status
ICH Q2(R2)	Validation of Analytical Procedures	✔ Applied
ICH Q3A-Q3B	Impurity Testing	✔ Applied
ICH Q8(R2)	Pharmaceutical Development	✔ Applied
ICH Q9	Quality Risk Management	✔ Applied
ICH Q14	Analytical Procedure Development	✔ Applied
USP <621>	Chromatography	✔ Applied
USP <1220>	Analytical Procedure Lifecycle	✔ Applied

Recommended Validation Parameters

Parameter	Acceptance Criteria	ICH Ref
Specificity	No interfering peaks at tR	Q2(R2) 4.1
Linearity	$R^2 > 0.999$ (0.1–100 µg/mL)	Q2(R2) 4.2
Accuracy	95–105% recovery	Q2(R2) 4.3
Precision	RSD < 2% (system), < 5% (method)	Q2(R2) 4.4
LOD / LOQ	S/N ≥ 3 / S/N ≥ 10	Q2(R2) 4.7
Robustness	Resolution maintained ±2% variation	Q2(R2) 4.6

Section 10 — Audit Metadata

Property	Value
Report ID	RPT-20260526151056-e3d836
Timestamp	2026-05-26T15:10:56.002007
Software Version	Chromatography AI System v3.0
RDKit Version	N/A
Processing Time	0.05 s
Status	Success
Overall Score	0.32

This report was generated by the CHROME-pred AI-Assisted Chromatographic Method Development System. All recommendations are based on peer-reviewed literature and should be validated experimentally for critical applications.